USGS
science for a changing world

U.S. Geological Survey—Reducing the Risk from Volcano Hazards

Lahar — River of Volcanic Mud and Debris

Lahar, an Indonesian word for volcanic mudflow, is a mixture of water, mud, and volcanic rock flowing swiftly along a channel draining a volcano. Lahars can form during or after eruptions, or even during periods of inactivity. They are among the greatest threats volcanoes pose to people and property. Lahars can occur with little to no warning, and may travel great distances at high speeds, destroying or burying everything in their paths.

Lahars form in many ways. They commonly occur when eruptions melt snow and ice on snow-clad volcanoes; when rains fall on steep slopes covered with fresh volcanic ash; when crater lakes, volcano glaciers or lakes dammed by volcanic debris suddenly release water; and when volcanic landslides evolve into flowing debris. Lahars are especially likely to occur at erupting or recently active volcanoes.

Because lahars are so hazardous, U.S. Geological Survey scientists pay them close attention. They study lahar deposits and limits of inundation, model flow behavior, develop lahar-hazard maps, and work with community leaders and governmental authorities to help them understand and minimize the risks of devastating lahars.

Lahars are among the most destructive and far-reaching ground hazards associated with volcanic eruptions. They can travel many tens of miles down valleys, move tens of miles per hour, and spill far beyond river-channel confines. Unlike floodwaters that recede and generally leave structures standing but coated in relatively thin deposits, lahars are slurries of sediment that can transport abundant rocks and woody debris. As a result, they can demolish buildings and other infrastructure and deposit muddy sand and gravel many feet (sometimes tens of feet) deep. Because of their destructiveness and propensity to deposit thick, rocky sediment, the effects of lahars can be long lasting, making reoccupation of inundated land difficult. Between 1600 and 2010, lahars caused more than 44,000 deaths worldwide, nearly 20 percent of all recorded fatalities associated with volcanic eruptions.

Lahars can occur during or after eruptions. Primary lahars are associated directly with eruptions. Secondary lahars occur after eruptions or during quiet periods between eruptions. Secondary lahars are a result of remobilization of volcanic sediment by heavy rainfalls, landslides, lake breaches, or water releases from glaciers, and they can occur weeks, months, or even centuries after an eruption.



Oblique aerial view of the remains of the town of Armero, Colombia, devastated by a lahar associated with the November 1985 eruption of Nevado del Ruiz. The lahar traveled more than 45 miles before destroying the town and killing more than 20,000 residents. U.S. Geological Survey photograph by R.J. Janda.

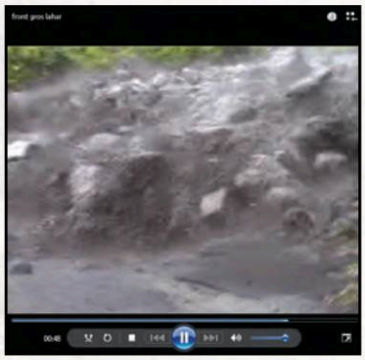
Eruptions, Landslides, Floods, and Downpours: Many Processes Generate Lahars

Lahars form in many ways. Major initiation mechanisms include:

- when hot volcanic rocks from explosive eruptions melt snow and glacier ice to form a mixture of water and debris that flows down slope;
- when very wet volcanic landslides liquefy and flow;
- when explosive eruptions eject water from crater lakes to form floods that erode sediment;
- when floods by other processes form at volcanoes and erode sediment from volcano slopes and river channels; and
- when rainfall erodes fresh volcanic ash from steep slopes.

How lahars begin can greatly influence flow volume, composition, behavior, and downstream damage. Lahars formed during explosive eruptions by snow-and-ice melt or ejection of crater lakes, by the failure of vast amounts of wet, weak rock from a volcano, or by sudden release of water from large lakes dammed by volcanic debris are likely to be large, fast, and very destructive far downstream from the volcano. Lahars formed by rainfall erosion of volcanic ash or by release of modest volumes of subglacial water are likely to be smaller and less far-travelled, but they are also more frequent and can overwhelm river channels and significantly affect downstream communities closer to volcanoes.

Video **of a small lahar at Curah Lengkong River, Semeru volcano, Indonesia, January 19, 2002. Flow front is 11.5 feet tall, and flow is moving at a speed of about 12 miles per hour. Video courtesy of F. Lavigne, University of Paris, used with permission.**



View video online at https://doi.org/10.3133/fs20183024.

Lahar—A Range of Sediment-Water Mixtures

A lahar can look and act very differently depending on its ratio of solids to water. A high-sediment-concentration *debris-flow lahar* has roughly equal proportions of sediment and water and looks very much like a flow of wet concrete. It transports sediment that typically ranges broadly in size, from clay particles to boulders as large as cars. A more dilute *hyperconcentrated-flow lahar* consists of more water than sediment, and has a more liquid appearance commonly described as oily or creamy. It is a type of flow broadly transitional between muddy water and a dense, concrete-like slurry. The sediment carried by this intermediate-type flow is dominantly sand, but it can roll large rocks along the channel bed, and it can be very damaging to channel banks and bridge piers. As a lahar moves downstream, its character and appearance can evolve as it erodes or deposits sediment and mixes with river water.





Oblique downstream views of a water flood (left) and high-sediment-concentration slurry (right) that formed on White Salmon River following the breach of Condit Dam, Washington. The high-concentration slurry is very similar to lahars that form at volcanoes. Note the thick, almost milkshake-like, texture of the slurry compared to the turbulent water flood. Inset images show sediment-concentration samples from the passing flows. Channel is about 100 feet wide. U.S. Geological Survey photographs by J.E. O'Connor.

Eruption-Triggered Melting and Mixing of Snow and Ice

At snow-covered and glaciated volcanoes, hot, dry flows of volcanic rock debris and gas—known as pyroclastic flows or surges (or more generally as pyroclastic currents)—commonly trigger lahars. The hot currents sweep across snow- and ice-covered slopes, intensely scour and mix with snow and glacier ice, and produce watery floods or slurries that flow away from the volcano. As these volcanically triggered flows race downhill, they erode additional sediment from the volcano's flanks and river channels, grow in volume, and become sediment-rich lahars. Notable lahars of this type formed during eruptions of Mount St. Helens, Washington, in 1980, and Nevado del Ruiz, Colombia, in 1985.

At Mount St. Helens, hot debris from a pyroclastic current swept across the snow-covered upper west and east flanks of the volcano. On each flank, the hot debris scoured and mixed with snow and ice and swiftly formed lahars with volumes of about 20 million cubic yards. These lahars traveled many tens of miles downvalley at speeds of tens of miles per hour.

At Nevado del Ruiz, pyroclastic currents from a modest-sized eruption rapidly melted snow and ice on the volcano's summit and produced abundant meltwater. The resulting floods eroded ample sediment from several steep, narrowly confined valleys, and transformed to lahars. Those lahars descended more than 16,000 feet in elevation from the summit and traveled more than 60 miles. One lahar emerged from a confined canyon 45 miles downstream from the summit and destroyed the town of Armero, killing more than 20,000 people.

Eruptions of ice-clad Redoubt volcano, Alaska, in 1989–90 and 2009 also generated large lahars. Several of these lahars flowed more than 25 miles to Cook Inlet and threatened or partly damaged an important oil storage and transfer facility. In 2017, the terminal owner decided to close the facility to eliminate environmental risk from future lahars.





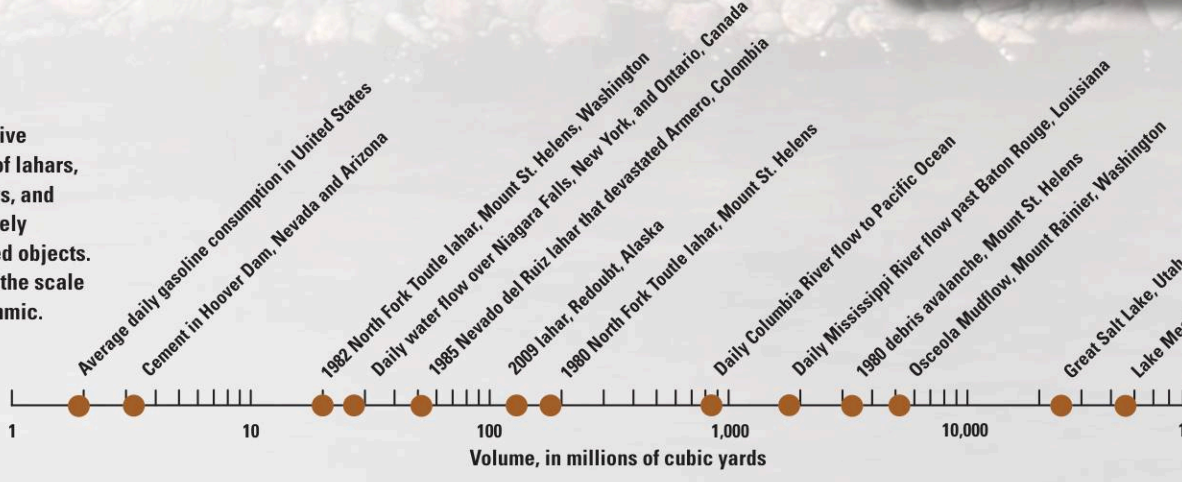


House and bridge (left) damaged and buried by lahars from the 1980 eruption of Mount St. Helens, Washington. U.S. Geological Survey photographs by L. J. Topinka (above) and R.B. Waitt (below).

Lahar flows can be deep. Mud coating on trees (above right) shows the depth of a lahar in Toutle River valley caused by the 1980 eruption of Mount St. Helens. U.S. Geological Survey photograph by L.J. Topinka.

Deposits of ancient (~2,500-year-old) lahars at Mount St. Helens, Washington (immediate right), caused by breaching of a lake dammed by volcanic material. Note the coarse, poorly sorted textures of the deposits that consist of particles ranging in size from microscopic clay to basketball-size boulders. U.S. Geological Survey photograph by J.J. Major.





Comparative volumes of lahars, river flows, and other widely recognized objects. Note that the scale is logarithmic.